

BUCKEYE, AZ, USA

WMO: 720644

Lat: 33.417N

Lon: 112.683W

Elev: 1021

StdP: 14.16

Time Zone: -7.00 (W07)

Period: 09-19

WBAN: 00226

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |            |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%        |            |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP         | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>12     | (b)<br>32.1 | (c)<br>36.0 | (d)<br>0.3                    | (e)<br>5.8 | (f)<br>55.6 | (g)<br>4.8 | (h)<br>7.3 | (i)<br>56.8 | (j)<br>19.8           | (k)<br>56.8 | (l)<br>17.7 | (m)<br>58.7 | (n)<br>4.9            | (o)<br>340 | (p)<br>0.449 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |              |             |              |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|--------------|-------------|--------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%           |             | 2%           |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB           | MCWB        | DB           | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>23.7            | (c)<br>109.1    | (d)<br>69.0 | (e)<br>108.0 | (f)<br>69.0 | (g)<br>104.4 | (h)<br>68.6 | (i)<br>76.3         | (j)<br>87.3 | (k)<br>75.1 | (l)<br>90.2 | (m)<br>74.1 | (n)<br>91.1 | (o)<br>8.2           | (p)<br>230 |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 74.5                            | 134.1 | 79.3 | 72.2 | 123.6 | 80.3 | 68.4 | 108.4 | 82.7 | 40.4          | 87.0 | 39.1 | 88.8 | 38.2 | 90.3 | 95.5              |

Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |           |                    |         | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |  |  |
|-------------------|----------|----------|----------------------------|-----------|--------------------|---------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|--|--|
|                   |          |          | Mean                       |           | Standard Deviation |         | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |  |  |
| 1%                | 2.5%     | 5%       | Min                        | Max       | Min                | Max     | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       |  |  |
| (a) 19.2          | (b) 16.7 | (c) 14.2 | (e) 27.3                   | (f) 114.3 | (g) 3.4            | (h) 2.4 | (i) 24.9                                           | (j) 116.0 | (k) 22.9   | (l) 117.4 | (m) 20.9   | (n) 118.8 | (o) 18.5   | (p) 120.5 |  |  |
|                   |          |          | DB                         |           |                    |         |                                                    |           |            |           |            |           |            |           |  |  |
|                   |          |          | WB                         |           |                    |         |                                                    |           |            |           |            |           |            |           |  |  |
|                   |          |          |                            |           |                    |         |                                                    |           |            |           |            |           |            |           |  |  |

Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 71.7   | 53.0  | 55.6  | 63.1  | 69.5  | 75.8  | 87.8  | 92.3  | 91.3  | 85.2  | 72.6  | 61.6  | 52.1  |
|                                                                        | DBStd       | 15.30  | 5.25  | 6.36  | 5.78  | 6.00  | 6.18  | 5.10  | 4.08  | 4.32  | 5.41  | 6.00  | 5.55  | 5.66  |
|                                                                        | HDD50       | 98     | 31    | 23    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 44    |
|                                                                        | HDD65       | 1313   | 373   | 266   | 104   | 25    | 3     | 0     | 0     | 0     | 0     | 11    | 130   | 401   |
|                                                                        | CDD50       | 8036   | 124   | 179   | 408   | 586   | 799   | 1135  | 1312  | 1279  | 1055  | 702   | 349   | 108   |
|                                                                        | CDD65       | 3773   | 0     | 2     | 46    | 161   | 336   | 685   | 847   | 814   | 605   | 248   | 29    | 0     |
|                                                                        | CDH74       | 57740  | 15    | 129   | 931   | 2364  | 4699  | 11001 | 13744 | 12893 | 8534  | 2912  | 509   | 9     |
|                                                                        | CDH80       | 36032  | 0     | 13    | 282   | 1034  | 2560  | 7429  | 9461  | 8652  | 5168  | 1325  | 108   | 0     |
| (14) Wind                                                              | WSAvg       | 6.1    | 5.6   | 5.7   | 6.0   | 7.0   | 6.9   | 6.3   | 6.7   | 6.2   | 6.2   | 5.9   | 5.6   | 5.5   |
| (15) Precipitation                                                     | PrecAvg     | 7.7    | 0.7   | 0.8   | 0.8   | 0.2   | 0.1   | 0.1   | 0.7   | 1.1   | 0.9   | 0.5   | 0.6   | 0.9   |
|                                                                        | PrecMax     | 14.5   | 4.3   | 3.0   | 3.3   | 1.8   | 1.3   | 1.5   | 2.4   | 4.3   | 4.6   | 3.1   | 2.8   | 4.7   |
|                                                                        | PrecMin     | 2.9    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
|                                                                        | PrecStd     | 3.3    | 0.8   | 0.8   | 0.8   | 0.4   | 0.3   | 0.2   | 0.7   | 1.0   | 1.1   | 0.7   | 0.7   | 1.2   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 75.4  | 81.9  | 91.0  | 96.8  | 103.6 | 113.2 | 111.3 | 111.3 | 107.7 | 97.4  | 86.3  | 75.1  |
|                                                                        |             | MCWB   | 53.0  | 54.9  | 58.7  | 59.9  | 63.0  | 67.6  | 70.4  | 69.8  | 68.9  | 63.6  | 57.6  | 56.8  |
|                                                                        | 2%          | DB     | 72.4  | 77.4  | 86.0  | 92.9  | 99.5  | 108.9 | 109.0 | 108.5 | 103.8 | 93.3  | 82.3  | 72.3  |
|                                                                        |             | MCWB   | 53.6  | 53.8  | 57.5  | 58.6  | 61.1  | 66.2  | 70.0  | 70.7  | 69.1  | 63.5  | 59.4  | 55.6  |
|                                                                        | 5%          | DB     | 69.7  | 74.7  | 82.3  | 90.2  | 95.4  | 106.4 | 107.8 | 106.2 | 101.6 | 90.9  | 80.8  | 68.1  |
|                                                                        |             | MCWB   | 50.9  | 52.9  | 55.2  | 57.7  | 60.0  | 66.1  | 69.8  | 70.6  | 68.2  | 63.0  | 58.4  | 52.7  |
|                                                                        | 10%         | DB     | 66.0  | 71.8  | 80.6  | 85.9  | 92.7  | 103.6 | 104.2 | 103.6 | 99.2  | 87.9  | 76.6  | 64.3  |
|                                                                        |             | MCWB   | 49.2  | 52.1  | 54.6  | 56.3  | 59.0  | 64.8  | 69.8  | 70.2  | 68.3  | 62.0  | 56.0  | 51.1  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 66.7  | 59.9  | 60.4  | 61.7  | 64.2  | 71.8  | 77.5  | 77.3  | 79.4  | 79.8  | 76.2  | 67.9  |
|                                                                        |             | MCDB   | 70.6  | 70.2  | 80.4  | 91.4  | 96.2  | 97.8  | 82.3  | 86.7  | 90.8  | 89.3  | 79.9  | 71.7  |
|                                                                        | 2%          | WB     | 57.9  | 57.5  | 59.0  | 59.8  | 62.6  | 70.4  | 75.9  | 76.0  | 74.8  | 70.1  | 63.2  | 59.4  |
|                                                                        |             | MCDB   | 60.9  | 68.0  | 78.8  | 88.3  | 93.8  | 97.1  | 87.7  | 90.0  | 87.5  | 80.3  | 71.0  | 64.1  |
|                                                                        | 5%          | WB     | 54.8  | 55.5  | 57.2  | 58.6  | 61.3  | 68.7  | 74.8  | 75.1  | 73.3  | 66.2  | 59.3  | 56.7  |
|                                                                        |             | MCDB   | 59.5  | 68.1  | 77.6  | 85.8  | 92.2  | 98.4  | 90.4  | 91.4  | 88.1  | 82.9  | 73.5  | 63.0  |
|                                                                        | 10%         | WB     | 52.2  | 53.7  | 55.6  | 57.3  | 60.0  | 66.8  | 73.7  | 74.2  | 72.1  | 64.0  | 57.1  | 53.8  |
|                                                                        |             | MCDB   | 62.0  | 67.3  | 76.2  | 83.2  | 89.9  | 99.3  | 92.3  | 91.9  | 88.7  | 81.2  | 74.1  | 61.3  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 25.4  | 26.5  | 29.0  | 29.7  | 28.9  | 30.6  | 23.7  | 22.9  | 24.8  | 27.3  | 27.5  | 23.8  |
|                                                                        | 5% DB       | MCDBR  | 30.7  | 32.5  | 33.7  | 34.0  | 32.9  | 31.9  | 26.1  | 26.0  | 28.1  | 30.7  | 31.0  | 28.4  |
|                                                                        |             | MCWBR  | 16.8  | 16.0  | 14.4  | 15.3  | 14.3  | 12.2  | 8.3   | 8.5   | 11.0  | 15.9  | 18.5  | 17.4  |
|                                                                        | 5% WB       | MCDBR  | 20.4  | 26.9  | 29.7  | 30.7  | 30.6  | 26.3  | 21.3  | 20.7  | 20.1  | 24.3  | 26.9  | 23.2  |
|                                                                        |             | MCWBR  | 14.5  | 13.4  | 12.4  | 13.6  | 12.8  | 10.0  | 6.0   | 6.5   | 8.6   | 16.6  | 20.3  | 19.9  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.277 | 0.282 | 0.304 | 0.324 | 0.335 | 0.342 | 0.400 | 0.389 | 0.346 | 0.313 | 0.285 | 0.277 |
|                                                                        | taud        |        | 2.576 | 2.572 | 2.481 | 2.409 | 2.407 | 2.389 | 2.355 | 2.375 | 2.466 | 2.518 | 2.557 | 2.580 |
|                                                                        | Ebn at Noon |        | 302   | 311   | 310   | 305   | 301   | 297   | 280   | 282   | 291   | 295   | 296   | 296   |
|                                                                        | Edh at Noon |        | 25    | 28    | 33    | 37    | 38    | 38    | 39    | 38    | 33    | 29    | 25    | 23    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1086  | 1394  | 1862  | 2311  | 2585  | 2677  | 2362  | 2172  | 1949  | 1568  | 1197  | 983   |
|                                                                        | RadStd      |        | 72    | 99    | 79    | 60    | 74    | 95    | 100   | 70    | 67    | 80    | 46    | 50    |

Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (12 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +194  | N/A   |

Nomenclature: See separate page